PACKET SMOKE AI SYSTEM

Startup and Main Function Initialization

Sanadama Hojii Seensaa, Sirna Jalqabaa (Startup Sequence) fi Dalagaa Ijoo (Main Function Architecture) Projeektii Packet Smoke AI

Gosa Sanadaa: Guula Tooftaa fi Injinaringii Toora Komputaraa

Mata Duree: Startup Events, Lifespan Handlers, Main Entrypoint Configuration

Qaama Xiyyeeffannoo: Injinoota Saayinsii AI, Deevoolooparoota Backend, fi Injinoota Toora Internetii

Haala Sanadaa: Mirkanaa'aa / Approved Standard

1. Seensa Projeektii Packet Smoke AI fi Jalqabbii Sirnaa

Projeektiin **Packet Smoke AI** sirna adda duree saayinsii dataa fi barnoota maashinii (Machine Learning) fayyadamuun dambali'a paakeetii toora intarnetii (network traffic flow) fi aara kusaana dataa (data smoke/anomaly detection) xiinxaluudha. Sirni kun hojii isaanii yeroo eegalu (startup) tajaajiloota baay'ee walxaxaa ta'an yeroo tokkotti qindeessuu gaafata.

1.1 Kaayyoo Jalqabbii (Startup Architecture)

Hojii ijoo (main function) fi tartiiba jalqabbii (startup sequence) seexaan qindeessuun akka sirni Packet Smoke AI yeroo eegalu doogaan (errors) akka hin uumamneef fi tajaajiloonni haala sirriin walqabataniif murteessaadha.

Gurguddoowwan Faayidaa Tartiiba Startup:

- **Walqabiinsa Kuusaa Dataa (Database Connections):** Walqabiinsa PostgreSQL fi Redis yeroo eegaleetti mirkaneessuu.
- **Fe'iinsa Modela AI (AI Model Loading):** Modela barnoota maashinii yeroo jalqabaa gara yaadannoo (RAM/GPU) fe'uu.
- **Qaaccessa Toora Intarnetii (Packet Sniffer Engine):** Tajaajila paakeetii tooraa kuffisu fi sakkata'u eagalchiisuu.

2. Caasaa Dalagaa Ijoo (Main Function Architecture)

Faayiliin `main.py` yookaan dalagaan `main()` bakka seensaa (entrypoint) sirna Packet Smoke AI ti. Inniis tajaajila FastAPI, tajaajiloota duubaa (background workers), fi AI pipelines walitti qabsiisa.

2.1 Caasaa Kodii main.py

```
import uvicorn
from fastapi import FastAPI
from app.core.config import settings
from app.api.v1 import router as api_v1_router
from app.core.lifespan import lifespan

def create_application() -> FastAPI:
    app = FastAPI(
        title="Packet Smoke AI API",
        version="1.0.0",
        lifespan=lifespan
    )

    # Routing galmeessuu
    app.include_router(api_v1_router, prefix="/api/v1")

    return app

app = create_application()

if __name__ == "__main__":
    uvicorn.run(
        "main:app",
        host="0.0.0.0",
        port=8000,
        reload=settings.DEBUG
    )
```

3. Tajaajila Lifespan Handlers (Startup fi Shutdown)

FastAPI keessatti toftaani harawaa `lifespan` jedhamu tartiiba jalqabbii (startup) fi cufamiinsa (shutdown) haala gaariin to'achuuf gargaara.

3.1 Kodii Lifespan Manager

```
from contextlib import asynccontextmanager
from fastapi import FastAPI
from app.services.ai_model import load_smoke_model, unload_smoke_model
from app.db.session import init_db, close_db

@asynccontextmanager
async def lifespan(app: FastAPI):
    # --- STARTUP EVENT ---
    print("Packet Smoke AI: Sirni eegalaa jira...")
    await init_db()
    app.state.ai_model = load_smoke_model()
    print("Modela AI fi DB haala gaariin fe'ameera.")

    yield  # Sirni asirratti hojii isaa geggeessa

    # --- SHUTDOWN EVENT ---
    print("Packet Smoke AI: Sirni cufamaa jira...")
    unload_smoke_model()
    await close_db()
    print("Tajaajiloonni cufamanii jiru.")
```

4. Modela AI Yeroo Startup Fe'uu (AI Model Preloading)

Modela barnoota maashinii (Machine Learning/Deep Learning) yeroo kaffaltiin (request) dhufu fe'uun sa'aatii dheeraa fudhata. Kanaafuu, yeroo sirni eegalu (startup) gubbaatti modela RAM keessatti olkaahuun barbachisaadha.

4.1 Caasaa Fe'iinsa Modela Packet Smoke

```
import torch

class SmokeDetectionAI:
    def __init__(self):
        self.model = None

    def load_model(self, model_path: str):
        print(f"Modela fe'aa jira: {model_path}")
        # PyTorch modela fe'uu
        self.model = torch.load(model_path)
        self.model.eval()
        return self.model

ai_engine = SmokeDetectionAI()
```

Kufaatii Alaa Ittisuu (Error Handling):

Yoo modeliin AI yeroo startup fe'amuu dadhabe, sirni sun hatattamaan dhaabbatee gabaasa (log) kaffaluu qaba.

5. Jalqabbii Qaaccessa Paakeetii (Packet Sniffer Integration)

Packet Smoke AI'n paakeetii toora intarnetii sakkata'uuf tajaajila duubatti hojjetu (background task) eegalchiisa.

5.1 Threading/Asyncio Bakkalcha Jalqabuu

```
import asyncio
from scapy.all import sniff

def packet_callback(packet):
    # Packet Smoke AI anomaly detection
    pass

async def start_packet_sniffer():
    print("Packet Sniffer eegalee jira...")
    loop = asyncio.get_event_loop()
    await loop.run_in_executor(None, sniff, prn=packet_callback, store=0)
```

6. Qindeessaa fi Sirna Naannoo (Environment Configurations)

Gubbaa main function fi startup irratti qindeessawwan (settings) adda addaa Pydantic BaseSettings fayyadamuun fe'amu.

6.1 Pydantic Settings Class

```
from pydantic_settings import BaseSettings

class Settings(BaseSettings):
    PROJECT_NAME: str = "Packet Smoke AI"
    DATABASE_URL: str
    AI_MODEL_PATH: str = "models/smoke_v1.pt"
    DEBUG: bool = False

    class Config:
        env_file = ".env"

settings = Settings()
```

7. Galmee Sirnaa fi Hordoffii (Logging & Monitoring)

Yeroo sirni eegalu galmeewwan (logs) gahaa ta'an uumuun rakkoo (bugs) yeroo jalqabaa fashaleessuuf gargaara.

```
import logging

logging.basicConfig(
    level=logging.INFO,
    format="%(asctime)s - %(name)s - %(levelname)s - %(message)s"
)
logger = logging.getLogger("PacketSmokeAI")

logger.info("Startup Event: Tajaajilli Packet Smoke AI eegaleera.")
```

8. To'annoo Doogaa Yeroo Startup (Exception Handling)

Yoo walqabiinsi DB fashalaaye yookaan modeliin AI badde, sirni kun akka inni sirriitti dhaabbatu (graceful degradation) gochuun barbaachisaadha.

Tarkiifama Yeroo Dogoggoraa:

- Log gochuu fi ergaa gabaasaa gara Sentry/Datadog erguu.
- Hojii sirnaa dhaabuu (Sys.exit(1)) akka container'n kufu gochuuf.

9. Guula Maayii fi Tooftaawwan Gaarii (Best Practices)

Caasaan startup fi main function qulqulluu fi unka sirrii qabu projeektii Packet Smoke AI gara fuulduraatti bal'isuuf fi hojii isaa saffisiisuuf murteessaadha.

9.1 Gabatee Muuxannoo Gaarii (Best Practices Matrix)

Tarkaanfii	Subii/Toofta Soba	Toofta Gaarii (Recommended)
AI Preloading	Kaffaltii (Request) irratti fe'uu	Startup event irratti RAM keessa kaa'uu
Lifespan	Deprecated `@app.on_event` fayyadamuu	Async `@asynccontextmanager` lifespan fayyadamuu
Config	Variables hardcode gochuu	Pydantic BaseSettings `.env` fayyadamuu

> **"Startup fi Main function sirrii ta'e amanamummaa fi saffisa projeektii Packet Smoke AI mirkaneessa."**